Higher National Diploma in Information Technology

Second Year First Semester Examination – 2021

HNDIT 2312-Principles of Software Engineering

Instructions

Answer Any four (04) Questions

All Question carry equal marks

No. of question: 05

No of pages: 02

Time: Two (02) hours

Q1)

I. What is software engineering? (04 marks)

II. Write down four application types. (04 marks)

III. What are the attributes of good software? (05 marks)

IV. Briefly explain the cost of software engineering. (06 marks)

V. Briefly explain the key challenges facing software engineering. (06 marks)

(Total 25 marks)

Q2)

I. What are the four activities of the software process? (04 marks)

II. Write down four agile software development methods. (04 marks)

III. Briefly explain the benefits of software prototyping. (05 marks)

IV. Briefly explain the spiral model with a suitable diagram. (06 marks)

V. Briefly explain the rapid application development industry. (06 marks)

(Total 25 marks)

Q3)

I. What are the main steps of requirement engineering? (04 marks)

II. Compare and contrast functional and nonfunctional requirements. (04 marks)

III. Define the term called stakeholders. Give 02 examples. (05 marks)

IV. Briefly explain the methods of requirements discovery. (06 marks)

V. Briefly explain the best practice of user interface design. (06 marks)

(Total 25 marks)

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Q4)

I. Define the following architectural patterns. (04 marks)

a. Layered architecture

b. MVC

II. What is cohesion? Give 02 types of cohesion. (04 marks)

III. Suggest an architecture pattern for a system (such as iTunes) that is used to sell and distribute music on the Internet. Justify your answer with the diagram. (05 marks)

IV. Briefly explain the roles of software quality assurance (QA) engineer. (06 marks)

V. Briefly explain following software testing methods. (06 marks)

a. Unit testing

b. Integration testing

c. Beta testing

(Total 25 marks)

Q5)

I. What are the reasons for software maintenance? (04 marks)

II. What is the version control system? Give 02 examples. (04 marks)

III. Briefly explain the concepts of system building. (05 marks)

IV. Briefly explain the activities of software project management. (06 marks)

V. Suppose you have a list of tasks shown as below. Draw the network diagram and find the critical path. (06 marks)

Task ID	Task Description	Task Predecessors	Task Duration (hours)
A	Project start		0
B	Buy materials for A	A	10
C	Buy materials for B	A	20
D	Build A	B, C	30
E	Build B	B, C	20
F	Polish and finish B	E	40
G	Join A and B	D, F	20
H	Project finish	G	0

(Total 25 marks)

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